



Problem solving with standard form

Task A: Not quite standard form numbers

1) Write the following in correct standard form:

a) $398 \times 10^{34} = 3.98 \times 10^{36}$

b) $9000 \times 10^{24} = 9 \times 10^{27}$

c) $340 \times 10^9 = 3.4 \times 10^{11}$

d) $89\,000 \times 10^{18} = 8.9 \times 10^{22}$

e) $42 \times 10^{-5} = 4.2 \times 10^{-4}$

f) $490\,000 \times 10^{-12} = 4.9 \times 10^{-7}$

g) $99 \times 10^{-2} = 9.9 \times 10^{-1}$

2) Write the following in correct standard form:

a) $0.034 \times 10^{39} = 3.4 \times 10^{37}$

b) $0.0056 \times 10^{23} = 5.6 \times 10^{20}$

c) $0.000\,032 \times 10^9 = 3.2 \times 10^4$

d) $0.45 \times 10^5 = 4.5 \times 10^4$

e) $0.583 \times 10^{-2} = 5.83 \times 10^{-3}$

f) $0.003\,53 \times 10^{-3} = 3.53 \times 10^{-6}$

g) $0.1 \times 10^1 = 1 \times 10^0$

3) Write the following in correct standard form:

a) $3590 \times 10^9 = 3.59 \times 10^{12}$

b) $0.003\,59 \times 10^9 = 3.59 \times 10^6$

c) $35.9 \times 10^{-9} = 3.59 \times 10^{-8}$

d) $0.359 \times 10^{-9} = 3.59 \times 10^{-10}$

e) $359 \times 10^{-2} = 3.59 \times 10^0$

f) $359 \times 10^2 = 3.59 \times 10^4$

g) $35.9 \times 10^{-2} = 3.59 \times 10^{-1}$



Task B: Using standard form in problems

1) How many years are 1 billion seconds?

Use a calculator and use 365 days as a standard year.

$$\begin{aligned}
 1 \text{ billion} &= 1 \times 10^9 = 1\,000\,000\,000 \text{ seconds} \\
 1\,000\,000\,000 \div 60 &= 16\,666\,666.7 \text{ (1 d.p.) minutes} \\
 16\,666\,666.7 \div 60 &= 277\,777.8 \text{ (1 d.p.) hours} \\
 277\,777.8 \div 24 &= 11\,574.1 \text{ (1 d.p.) days} \\
 11\,574.1 \div 365 &= 31.7 \text{ (1 d.p.) years}
 \end{aligned}$$

2) Flash drives come in 16 MB, 64 MB and 128 MB. A video was made and has 21 700 000 bytes.

Given that 1×10^6 is 1 MB (megabyte).

What is the minimum flash drive needed to put this film on?

$$\begin{aligned}
 21\,700\,000 &= 2.17 \times 10^7 \\
 \text{This is the same as } &21.7 \times 10^6 \\
 \text{Given we know } \times 10^6 &\text{ is the same as Mega we can say that } 2.17 \times 10^7 = 21.7\text{MB} \\
 \text{The minimum is a } &64\text{MB flash drive.}
 \end{aligned}$$

3) The radius of Mercury is approximately 2 500 km.

The radius of Neptune is approximately 25 000 000 m.

How many times bigger is Neptune's radius than Mercury's?

$$\text{Mercury's radius} \approx 2\,500 \text{ km} = 2\,500\,000 \text{ m} = 2.5 \times 10^6 \text{ m}$$

$$\text{Neptune's radius} \approx 2.5 \times 10^7 \text{ m}$$

Neptune's radius is 10 times bigger than Mercury's radius.

Planet	≈ Radius
Mercury	2 500 km
Venus	6 000 km
Earth	6 000 km
Mars	3 000 km
Jupiter	71 000 km
Saturn	60 000 km
Uranus	25 000 km
Neptune	25 000 km

4) Here are some approximations of the radii of planets.

Write any two of your own questions, including the use of standard form.

- Write all of these radii in m using standard form.
- Put the planets in ascending order, using standard form.

